

Murine antibody humanization via
frequentist analysis, simulated
annealing, and support vector
machine algorithm

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1 Intro

The aim of this GitHub repository is to delve deeper into biophysics, using simulated annealing and science data to humanize murine antibodies.

We use a frequentist analysis of the given data in `learn_human.txt` and `learn_mouse.txt` in order to measure entropies and correlations.

With those outputs we intend to fine-tune the simulated annealing algorithm for the humanization.

1.1 Raw data files

The `learn_human.txt` and `learn_mouse.txt` data entries are chains of 298 amino acids each. Every amino acid is represented by a character in this string: `-ACDEFGHIKLMNPQRSTVWY`. The `-` is a gap in the chain, as used in the notation convention.

1.2 amino acid's properties

Every amino acid is a certain chemical compound. Thus, it has chemical properties, which vary from amino acid to amino acid. The used properties in this project are (in order): property-gap, hydrophobic, aromatic, aliphatic, polar, small, minuscule, positively charged, and negatively charged. Every amino acid has assigned at least one property, so for consistency we added the property version of the gap: property-gap, denoted as `-(PROP)`.

2 Representation and modelization of data

2.1 amino acids and properties

Every amino acid will be simulated/coded as a 21-dimensional vector of real numbers between 0 and 1 juxtaposed to a 9-dimensional vector of real numbers between 1 and 0. That is, some amino acid in our amino acid space, $a \in \mathcal{A}$, will be $a = v \times w \in [0, 1]^{21} \times [0, 1]^9 = \mathcal{A} \subset \mathbb{R}^{21} \times \mathbb{R}^9$.

- The first list will be a 21-dimensional vector, whose i -th component corresponds to the frequency of the i -th amino acid from the aforementioned amino acid string.
- The second list will be a 9-dimensional vector, whose i -th component corresponds to the frequency of the i -th property from the aforementioned property list.

As you can see, these variables are context-dependent, since there is no reference for the frequency of amino acids or properties. Therefore, if it is one that represents an actual amino acid, all frequencies may be 0 but one of them, which should be 1. Same will happen for the property vector. We will call this a *concrete amino acid*.

Example: Let's encode the amino acid cysteine (`C`). It is the third in the amino acid list, thus, the first vector should be $v = (0, 0, 1, 0, \dots, 0)$. Cysteine is hydrophobic and small, then: $w = (0, 1, 0, 0, 0, 1, 0, 0, 0)$, since hydrophobic is in the 2nd position and small in the 6th.

Now changing the context, another setting we will be using is the so called *Schrödinger's amino acid*. That will be the average amino acid, whose elements in both vectors will be *separately* the frequencies of every appearing amino acid.

Separately means that the sum of the components of the first list is 1 and the sum of the components of the second list will also be 1.^[1] More specifically: $\sum_{i=1}^{21} v^i = 1 \wedge \sum_{i=1}^9 w^i = 1$.^[2]

¹This is true for every kind of amino acid, whether it is concrete or Schrödinger's.

²Using superscript for convenience.

The norm of an amino acid, $a = v \times w \in \mathcal{A}$, is defined as:

$$\|a\| = \sqrt{\sum_{i=1}^{21} (v^i)^2}$$

When treating with concrete amino acids, we know that $\|a\| = 1$, since only one of the components can be 1.

2.2 Chains

A chain is an array of 298 amino acids and it represents a whole antibody. Just as before, we will have concrete (all components must be concrete amino acids) and Schrödinger's chains (the rest).

This is a $298 \times (21 + 9)$ matrix of split row conditioned frequencies. We will name the space of all possible chains the chain space, $\mathcal{C} = \mathcal{A}^{298}$. Thus, $c \in \mathcal{C}$ will have the form $c = (c_1, \dots, c_j, \dots, c_{298})$, where $c_j = v_j \times w_j$ and $v_j = (v_j^1, \dots, v_j^i, \dots, v_j^{21})$, $\forall j \in [1, 298] \subset \mathbb{N}$.

The norm of a chain, $c \in \mathcal{C}$, is defined as:

$$\|c\| = \sqrt{\sum_{i=1}^{298} \|c_i\|^2}$$

If c is a concrete chain, then $\|c\| = \sqrt{298}$.

3 Expressions and Formulas

3.1 Entropies

We quantify the variability of the amino acids in a Schrödinger's chain with entropies, applying its expression to both vectors of every amino acid in that chain. This means the result will be a 298 element list of pairs of entropies.

To compute them as follows:

$$S_{v,j} = - \sum_{i=1}^{21} v_j^i \log_2 v_j^i, \quad S_{w,j} = - \sum_{i=1}^9 w_j^i \log_2 w_j^i, \quad S_j = (S_{v,j}, S_{w,j}) \quad \forall j \in [1, 298] \subset \mathbb{N}$$

That previous one is the Shannon entropy, but we may need to use some others:

- Linear entropy: $s = \sum_i p_i(1 - p_i)$
- Rényi entropy: $s_\alpha = \frac{1}{1-\alpha} \sum_i p_i^\alpha$, $\alpha \in [0, 1)$
- Tsallis entropy: $s_\alpha = \frac{1}{q-1} (1 - \sum_i p_i^\alpha)$, $\alpha \in [0, 1)$

3.2 Correlation

The measure of correlation between positions in a chain used in this project is mutual information. It is an entropy measure of how much information (in bits) we can deduce of one random variable given another one, and vice versa.

Mutual information is given by:

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P(\{X = x\} \cap \{Y = y\}) \log_2 \frac{P(\{X = x\} \cap \{Y = y\})}{P(\{X = x\})P(\{Y = y\})},$$

where X and Y are discrete random variables that take values, x and y , in $\mathcal{X}$ and $\mathcal{Y}$. Since we can identify our vectors, v and w , with distribution functions, we are able to apply this method.

Note that $I(X; Y) = I(Y; X)$, since the product and intersection are symmetric.

Also,

$$I(X; X) = \sum_{x \in \mathcal{X}} P(\{X = x\}) \log_2 \frac{P(\{X = x\})}{(P(\{X = x\}))^2} = - \sum_{x \in \mathcal{X}} P(\{X = x\}) \log_2 P(\{X = x\}) = S(X),$$

where $S(X)$ is the entropy of the random variable X .

In practice, we will get a 298×298 symmetric matrix for amino acids (v -part of the chain) whose diagonal is the entropies of every position. However, for our correlation indices we would rather have all 1's in the diagonal and the other elements going from 0 to 1. We normalize it as follows:

$$i(X; Y) = \frac{I(X; Y)}{\min\{S(X), S(Y)\}}.$$

When $\min\{S(X), S(Y)\} = 0$, the position is fully conserved and $I(X; Y) = 0$ as well. In this case, the code outputs $i(X; Y) \equiv 0$, by convention.

4 Metropolis

Important information: Metropolis assumes CDRs are located in the same position for all antibodies being analyzed; in order for this to be valid, sequences need to be properly aligned beforehand.

4.1 Simulation Workflow: From Batch to Single Run

Two functions are used for running the simulation: a high-level manager (`mega_metropolis`), and a core simulation engine (`run_metropolis`). This design allows for parallelized execution of simulations on multiple initial murine sequences.

4.1.1 The `mega_metropolis` Manager

`mega_metropolis`'s primary responsibilities are:

1. **Data Loading:** It begins by reading a list of initial murine sequences (which we call seeds) from a specified file and saves them for later use.
2. **Batch Management:** It creates a timestamped main directory for the entire simulation batch (format `results/metropolis/YYYY-MM-DD_HH_MM_SS..../`). .
3. **Parallel Execution:** It iterates through each seed sequence using an OpenMP parallel loop. For each seed, it creates a dedicated subdirectory (e.g., `seq_1`, `seq_2`).
4. **Replication:** For each seed, it calls the `run_metropolis` function `n_metropolis` times. This allows for multiple independent simulation runs starting from the same seed, providing statistical significance to the results. The output of each run is saved to a separate file (e.g., `run_1.txt`, `run_2.txt`) within the corresponding seed's subdirectory.

4.1.2 The `run_metropolis` Engine

The `run_metropolis` function performs a single, complete simulated annealing run on one given sequence:

1. **Initialization:** It takes an initial sequence, the human Schrödinger chain, and the pre-calculated matrix of β values.

2. **Simulated Annealing Loop:** It iterates through each row of the β matrix, representing a step in the “cooling” schedule.
3. **Metropolis Sweeps:** For each β^i step, it performs a specified number of Metropolis sweeps (`n_sweeps`). A single sweep involves attempting a mutation at every position in the sequence.
4. **State Acceptance:** We begin by proposing a new state, selecting a new AA (including the gap character) uniformly from the 21 possibilities (ensuring it’s different from the current one). Each proposed mutation is accepted or rejected based on the change in total energy (ΔE) and the current, position-specific, β_j^i value, following the Metropolis criterion: accept if $\Delta E < 0$ or if a random number is less than $e^{-\beta_j^i \Delta E}$.
5. **Data Logging:** After the simulation is complete, it calls a dedicated function to write a detailed report file containing the initial parameters, the final sequence and energy for each β step, and overall acceptance statistics.

4.2 Simulated Annealing Schedule (β)

The β -values, which are inverse of the temperature ($1/k_B T$), are not constant; instead, they are defined through a mechanism, which we call the “engine”:

- **Engine:**

$$\beta_i^j = k_i \cdot \frac{\alpha}{S_j + \varepsilon}, \quad k_i = (c_r)^i \quad \forall i \in [1, N_\beta], j \in [1, L_{\text{chain}}]$$

where c_r is the cooling rate, ε avoids division by 0, α is a scale factor, and S_j is entropy for position j in human reference Schrödinger chain.

4.3 Energy Function

The total energy, E_{total} , is a weighted sum of three components, which we aim to minimize:

$$E_{\text{total}} = w_{\text{log}} E_{\text{log}} + w_{\text{prop}} E_{\text{prop}} + w_{\text{penalty}} E_{\text{penalty}}$$

The weights for each component (`WEIGHT_LOG`, `WEIGHT_PROP`, `WEIGHT_PENALTY`) are defined in ‘head.h’.

1. **Log Humanness Energy, E_{log} :** This term quantifies how ”human-like” a sequence is based on AA frequencies from a reference set of human sequences (the Schrödinger chain):

$$E_{\text{log}} = - \sum_{i=1}^{L_{\text{chain}}} \log(p_i(a_i))$$

where $p_i(a_i)$ is the frequency of the amino acid a_i at position i in the human reference. A large penalty (`ZERO_FREQ_PENALTY_LOG`) is added for amino acids with a frequency below a small threshold (`EPSILON`).

2. **Property Distance Energy, E_{prop} :** This term measures the similarity between the properties of the sequence and the human reference; it’s calculated as the squared Euclidean distance in the property space for each position:

$$E_{\text{prop}} = \sum_{i=1}^{L_{\text{chain}}} \sum_{j=1}^{N_{\text{props}}} \left(\text{prop}_j(a_i) - \overline{\text{prop}_j(i)} \right)^2$$

where $\text{prop}_j(a_i)$ is the value of property j for amino acid a_i , and $\overline{\text{prop}_j(i)}$ is the average value of property j at position i in the Schrödinger chain.

3. **Wandering Penalty Energy, E_{penalty} :** This term discourages the sequence from deviating too far from the original murine sequence; we thus only incentivize mutation if there’s a relatively big gain in humanness. It consists of a penalty based on property distance and a (for now) constant penalty for any mutation:

$$E_{\text{penalty}} = \sum_{i=1}^{L_{\text{chain}}} \left[\left(\sum_{j=1}^{N_{\text{props}}} (\text{prop}_j(a_i) - \text{prop}_j(a_i^{\text{original}}))^2 \right) + P \cdot \delta_{a_i}^{a_i^{\text{original}}} \right]$$

where a_i^{original} is the amino acid at position i in the original murine sequence, P is the mutation penalty (AA_MUTATION_PENALTY), and δ is the “generalized Kronecker delta” (logical if gate). This approach ensures each mutation has a minimum energy cost, while heavily penalizing big property changes.

4.4 Model Parameters

The following table summarizes the key parameters for metropolis in our simulations. These values are defined in `main.c` and `head.h`.

Parameter	Value ^[3]	Description
<code>n_betas</code>	50	Number of cooling steps in the annealing schedule.
<code>n_sweeps</code>	800	Number of Metropolis sweeps per β step.
<code>n_metropolis</code>	20	Number of independent runs per murine seed sequence.
<code>w_log</code>	0.4 (<code>head.h</code>)	Weight for the log-humanness energy component.
<code>w_prop</code>	0.4 (<code>head.h</code>)	Weight for the property-distance energy component.
<code>w_penalty</code>	0.2 (<code>head.h</code>)	Weight for the wandering penalty energy component.
<code>cooling_rate</code> (c_r)	1.02	Multiplicative factor for increasing k_i at each step.
<code>scale_factor</code> (α)	1.0	Global scaling factor for all β values.
<code>AA_MUTATION_PENALTY</code> (P)	1.5 (<code>head.h</code>)	Constant penalty added for any AA mutation away from the original.

³Provisional values.

5 Support Vector Machine

We intend to back up the humanness of the chain outputted by the simulated annealing algorithm with this binary-output method.

The main point of an SVM (abbreviation of Support Vector Machine) is to transform our data space into some other one in such a way that we can separate the data points with a simple hyperplane, a linear classifier.

We will represent each point with a n -dimensional vector, where n is the number of data points. The function that gives us those coordinates is called the kernel.

5.1 The kernel

Our kernel, $k : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}$, will be the Gaussian radial basis function:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right), \quad \gamma > 0,$$

where $\{\mathbf{x}_i\}$ are the data points.

After applying the kernel we have a matrix $M \in \mathbb{R}^{n \times n}$, the Gram matrix, with the coordinates of our data points in the new space. That will be: $[M]_{i,j} = k(\mathbf{x}_i, \mathbf{x}_j)$.

We also have available four other standard kernel functions:^[4]

- Sigmoid function:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \tanh \frac{\langle \mathbf{x}_i | \mathbf{x}_j \rangle}{298}$$

- Linear function:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \frac{\langle \mathbf{x}_i | \mathbf{x}_j \rangle}{298}$$

- Tanimoto function:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \frac{\langle \mathbf{x}_i | \mathbf{x}_j \rangle}{596 - \langle \mathbf{x}_i | \mathbf{x}_j \rangle}$$

- Cauchy function:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \frac{298}{298 + \|\mathbf{x}_i - \mathbf{x}_j\|^2}$$

The Gram matrix and the kernel are the tools we need in order to craft the decision function.

5.2 Decision function

Using the lagrange multipliers method for minimization^[5], we get the decision function, $f : \mathcal{C} \rightarrow \mathbb{R}$:

$$f(\mathbf{x}) = b + \sum_{i=1}^n \lambda_i s_i k(\mathbf{x}_i, \mathbf{x}),$$

where λ_i are the lagrange multipliers obtained through said method, s_i is the *tag* of the i -th data point (may be +1, when human, or -1, when murine) and b is the bias, obtained as follows:

$$b = s_p - \sum_{i=1}^n \lambda_i s_i k(\mathbf{x}_i, \mathbf{x}_p),$$

where the p -th data point is a support one, that is, the closest to the plane (or any of them) and such that $\lambda_p > 0$.

⁴These constants (298 and $596 = 2 \cdot 298$) that are scattered around the functions seem to be arbitrary, but the written functions are already optimized for our use case: they are normalized and centered so that every output is positive and does not surpass 1.

⁵See [appendix for full derivation](#).

6 SVM results

Using the `learn_human`, `learn_mouse`, `test_human`, and `test_mouse` sets for training and testing, we get the following results. In this histogram, we show the distribution of the sign-corrected certainty degree (positive when correct and negative when failed tagging):

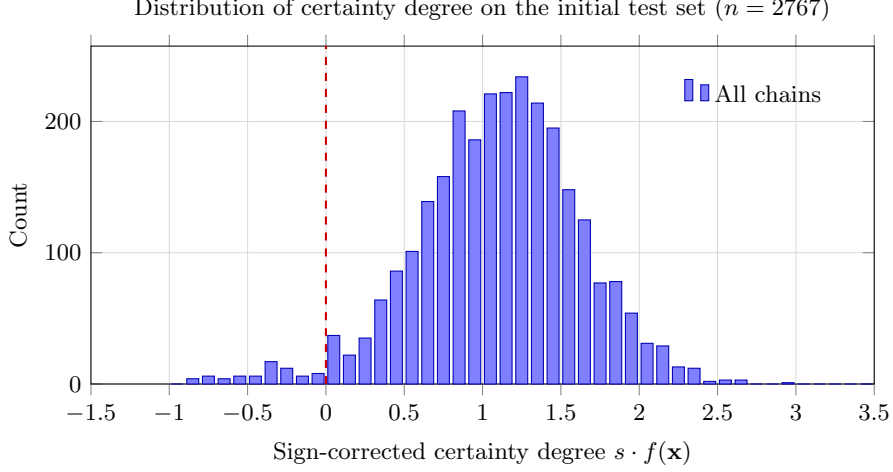


Figure 1: Count with respect to sign-corrected certainty degree on the initial dataset.

This gets us to the confusion matrix:

	Human	Murine	Overall percentages	Human	Murine
Tagged as human	1360	41	Tagged as human	49.15%	1.48%
Tagged as murine	28	1338	Tagged as murine	1.01%	48.36%

Counting human as positive, this gets us to a $F1^{[6]}$ of $\sim 0.9753^{[7]}$. On the other hand, being murine the positives, we have an $F1$ of $\sim 0.9749^{[8]}$. This takes us to an overall accuracy^[9] of $\sim 0.9751^{[10]}$.

However, those tests were taken with a completely heuristic upper limit, we will call λ_t , for the λ vector. Since we cannot determine λ_t analytically, we must adjust it as a hyperparameter with a validation data set.

⁶ $F1 = \frac{2 \cdot TP}{2 \cdot TP + FN + FP}$

⁷Precisely: $F1_H = 0.975259949802797$.

⁸Precisely: $F1_M = 0.974863387978142$.

⁹ $Acc = \frac{TP + TN}{TP + TN + FP + FN}$

¹⁰Precisely: $Acc = 0.975063245392121$.

6.1 Validation

We have merged all data sets and split them into **learn**, **validation**, and **test**, with 70%, 15%, and 15% of the chains respectively.

Through this method we get an optimal value for $\lambda_t = 10.0$, having tried with powers of 10 from 0.01 to 1000. Using that hyperparameter, we get the following confusion matrix

	Human	Murine	Overall percentages	Human	Murine
Tagged as human	404	4	Tagged as human	60.48%	0.60%
Tagged as murine	6	254	Tagged as murine	0.90%	38.02%

And F1 values and accuracy:

$$F1_H = 0.9878^{[11]} \quad F1_M = 0.9788^{[12]} \quad Acc = 0.9836^{[13]}$$

The following histogram shows the distribution of the sign-corrected certainty degree, as before positive when correct and negative when incorrect tagging:

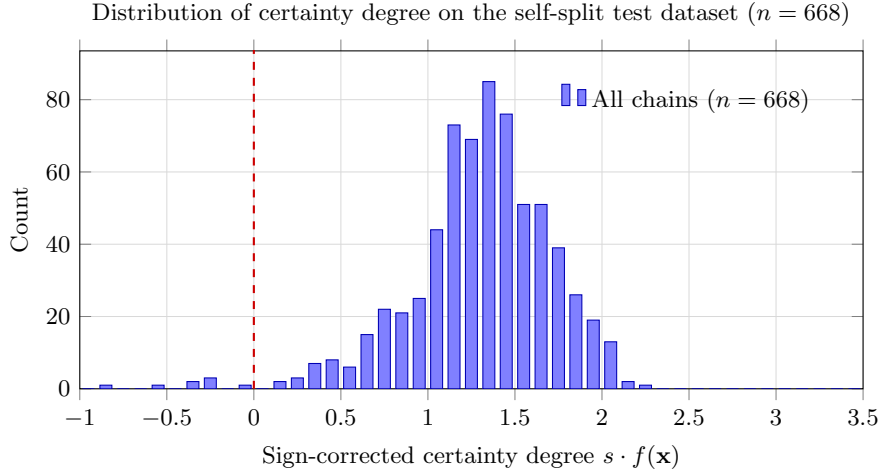


Figure 2: Count with respect to sign-corrected certainty degree using self-split learn, validation, and test datasets.

¹¹Precisely: $F1_H = 0.987775061124694$.

¹²Precisely: $F1_M = 0.978805394990366$.

¹³Precisely: $Acc = 0.98355754857997$

6.2 Cross-validation of standard kernels

And with every standard kernel we presented, we obtain similar results:

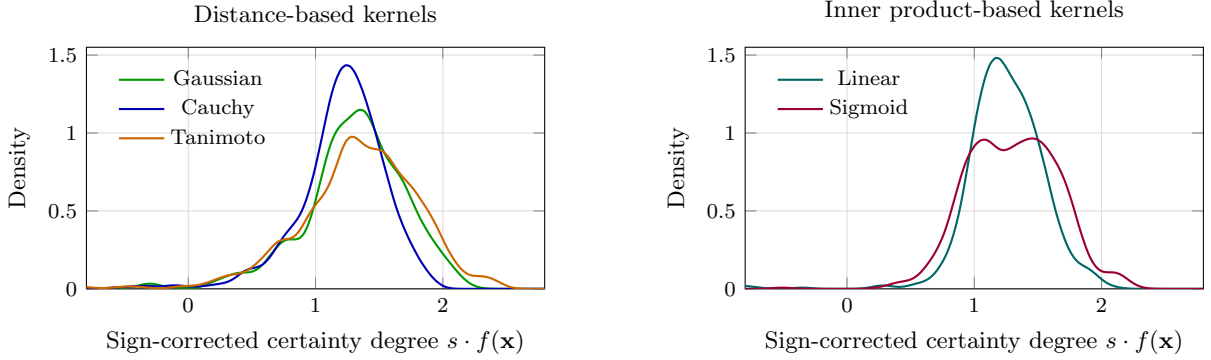


Figure 3: Density distributions of sign-corrected certainty degrees $s \cdot f(\mathbf{x})$ on the test dataset for distance-based kernels (left) and inner product-based kernels (right). Values greater than zero ($s \cdot f > 0$) are correct classifications, and values below zero ($s \cdot f < 0$) are errors.

To see how their predictions relate to each other, we plot the scatterplot matrix of decision function outputs $f(\mathbf{x})$ in Figure 4.

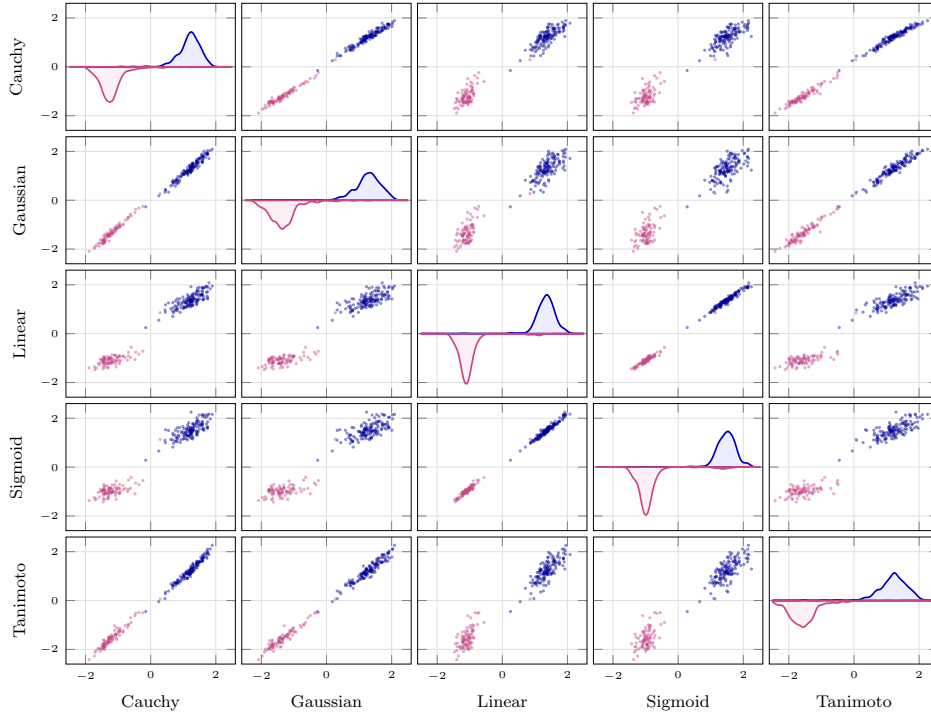


Figure 4: Scatterplot matrix of raw decision function outputs $f(\mathbf{x})$ for the five standard kernels. Off-diagonal panels show pairwise scatter distributions (human in blue, murine in pink). Diagonal panels display mirrored bilateral density estimates, with human density plotted upward ($y \geq 0$) and murine density mirrored downward ($y \leq 0$) to eliminate visual occlusion and maintain uniform axis scaling across the matrix.

Here we have the correlation matrix for the five standard kernels:^[14]

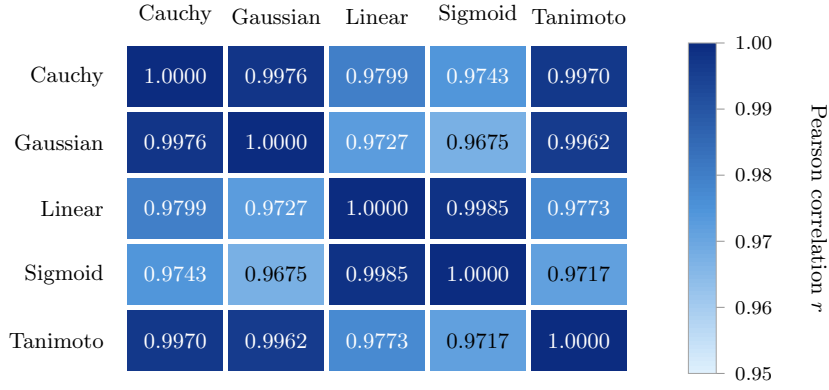


Figure 5: Correlation heatmap across the raw decision function outputs $f(\mathbf{x})$ of the five standard kernels on the test dataset.

As we can see from both the scatterplot matrix and the correlation heatmap, the five standard kernels are very correlated ($r > 0.96$ for all pairs, and $r \approx 0.998$ between Linear and Sigmoid). Because these functions are smooth across the whole range $[0, 1]$, they apply a uniform scale and do not focus on the narrow similarity interval where antibodies switch from murine to human.

¹⁴The correlations were expected, to a certain level, because of how similar the functions used as kernel are to each other. See appendix for detail.

6.3 Custom piecewise kernels

To focus on this boundary region, we define a piecewise linear kernel with a ramp of width W centered at C :

$$k(\mathbf{x}_i, \mathbf{x}_j) = \begin{cases} \frac{L}{C - \frac{W}{2}} x, & x < C - \frac{W}{2} \\ \frac{H - L}{W} \left(x - C + \frac{W}{2} \right) + L, & |x - C| \leq \frac{W}{2} \\ \frac{1 - H}{1 - C - \frac{W}{2}} \left(x - C - \frac{W}{2} \right) + H, & x > C + \frac{W}{2} \end{cases}$$

where $x = \frac{\langle \mathbf{x}_i | \mathbf{x}_j \rangle}{298} \in [0, 1]$, with baseline levels $L = 0.2$ and $H = 0.8$.

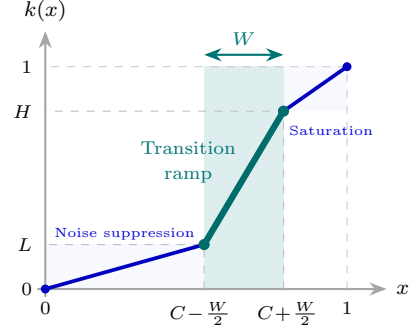


Figure 6: Mathematical definition and schematic mapping of the custom piecewise linear kernel. A narrow transition band of width W centered at C magnifies subtle sequence differences near the decision threshold ($\Delta k = \frac{H-L}{W} \Delta x \gg \Delta x$), while dissimilar ($x < C - W/2$) and highly similar ($x > C + W/2$) sequence pairs are compressed into muted baseline regions.

By choosing a narrow width ($W = 0.08$) and placing the center C in the transition region ($C \in \{0.78, 0.80, 0.82, 0.84\}$), small differences between sequences near the threshold produce large changes in the kernel, while sequences far from the boundary stay in constant regions.

To measure both classification accuracy and margin certainty in a single score between 0 and 1, we define the Quadratic Certainty Score $\mathcal{Q}$:

$$\mathcal{Q} = 1 - \frac{1}{n} \sum_{i=1}^n \left(1 - \frac{1}{1 + e^{-s_i f(\mathbf{x}_i)}} \right)^2 \in [0, 1]$$

where $s_i \in \{-1, +1\}$ is the true tag and $f(\mathbf{x}_i)$ is the decision function output. Correct predictions with high margin give values close to 1.0, predictions on the boundary give 0.75, and wrong predictions with high margin give values close to 0.0.

Evaluating these custom kernels on the test dataset gives the results shown in Figure 7 (see appendix for the full table):

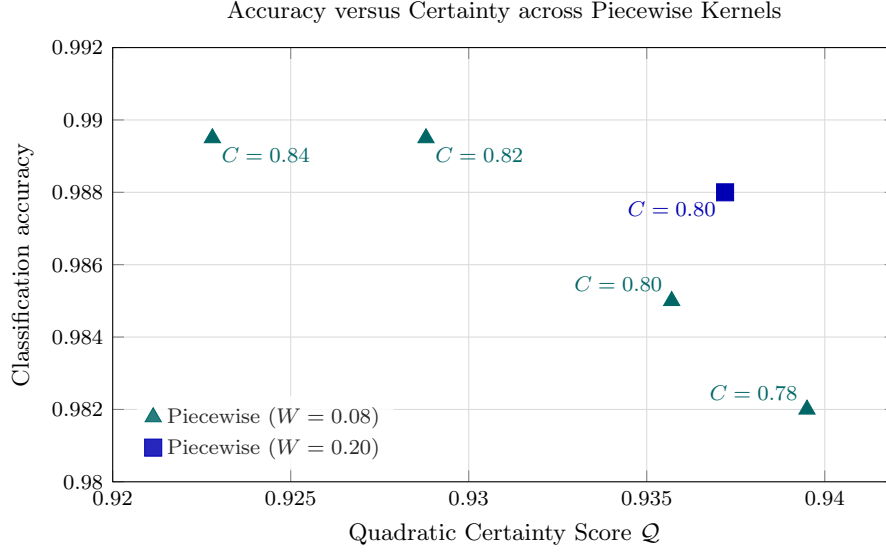


Figure 7: Scatterplot of classification accuracy versus Quadratic Certainty Score Q across the custom piecewise kernel configurations on the test dataset (see appendix for complete table).

The piecewise kernels with $C = 0.82$ and $C = 0.84$ reach an $F1_H$ of 0.9915 and 98.95% accuracy, higher than any of the five standard kernels.

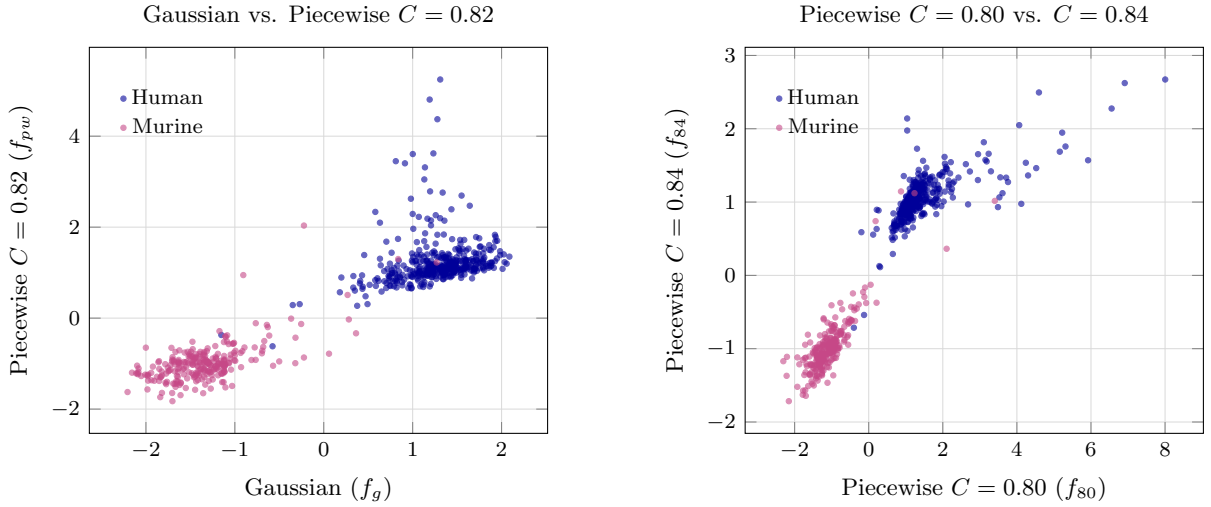


Figure 8: Scatterplots of decision output for custom piecewise kernels compared against standard Gaussian (left) and between two fine-tuned piecewise centers (right).

We now have 10 different kernel tests: 5 standard global kernels and 5 piecewise kernels. In the next section, we train a logistic regression model to find the best linear combination of all 10 kernel outputs.

7 Logistic regression

Now that we have computed the decision functions for several kernels, we can combine their results. Instead of taking only the sign of $f(\mathbf{x})$, we use the continuous values directly.

7.1 10-dimensional representation

For each antibody chain $\mathbf{x}$, we build a 10-dimensional vector using the output of the 10 SVM kernel tests:

$$\mathbf{v}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_{10}(\mathbf{x})) \in \mathbb{R}^{10},$$

where the components correspond to:

- f_1 : Cauchy
- f_2 : Gaussian (RBF)
- f_3 : Linear
- f_4 : Sigmoid
- f_5 : Tanimoto
- f_6 : Piecewise ($C = 0.80, W = 0.20$)
- f_7 : Piecewise ($C = 0.80, W = 0.08$)
- f_8 : Piecewise ($C = 0.82, W = 0.08$)
- f_9 : Piecewise ($C = 0.84, W = 0.08$)
- f_{10} : Piecewise ($C = 0.78, W = 0.08$)

Before fitting the model, we standardize each feature using the mean μ_k and standard deviation σ_k of the validation dataset:

$$\hat{f}_k(\mathbf{x}) = \frac{f_k(\mathbf{x}) - \mu_k}{\sigma_k}.$$

7.2 Model and training

We define the linear combination:

$$z(\mathbf{x}) = b + \sum_{k=1}^{10} w_k \hat{f}_k(\mathbf{x}),$$

where w_k is the weight of the k -th kernel and b is the bias. The probability of being human is given by the logistic function:

$$P(\text{Human} \mid \mathbf{x}) = \frac{1}{1 + e^{-z(\mathbf{x})}}.$$

If $P(\text{Human} \mid \mathbf{x}) \geq 0.5$, we classify the chain as human (+1), and as murine (−1) otherwise.

We train the model on the validation dataset (667 samples) using gradient descent to minimize the binary cross-entropy loss with L2 regularization:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \right] + \frac{\lambda}{2} \sum_{k=1}^{10} w_k^2,$$

where $y_i \in \{0, 1\}$, $\hat{y}_i = P(\text{Human} \mid \mathbf{x}_i)$, $\lambda = 0.01$, and learning rate $\eta = 0.5$ over 2000 iterations.

7.3 Weights and feature importance

After training, all weights are positive, meaning that a higher score in any kernel pushes the prediction towards human. In the following plot, we show the learned weights in descending order:

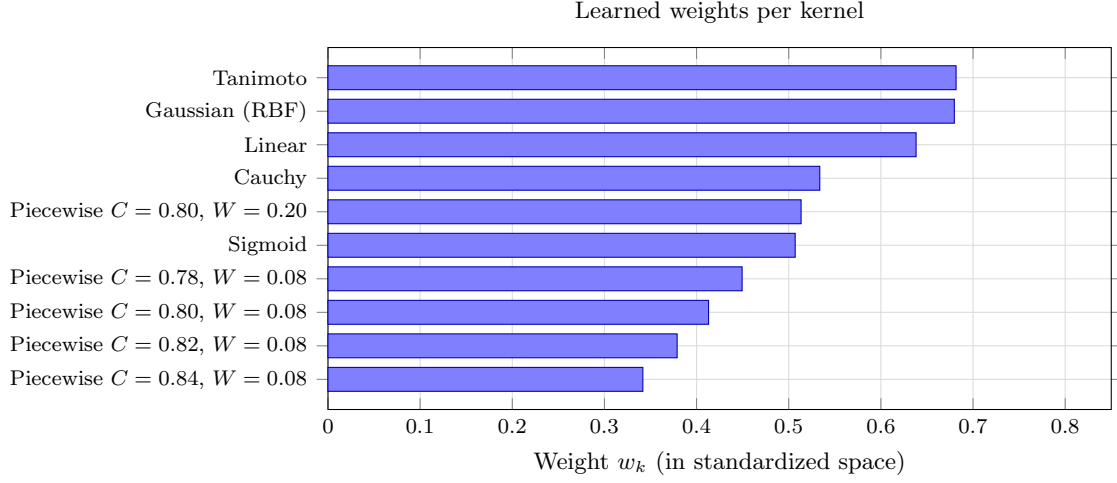


Figure 9: Learned logistic regression weights in descending order.

Tanimoto ($w \approx 0.6816$), Gaussian ($w \approx 0.6797$), and Linear ($w \approx 0.6382$) have the largest weights, while the narrow piecewise kernels ($W = 0.08$) add extra weight near the decision threshold.

7.4 Results

Evaluating the trained model on both datasets gives the following confusion matrices:

	Human	Murine
Tagged as human	403	2
Tagged as murine	2	260

Validation dataset (667 samples)

	Human	Murine
Tagged as human	403	4
Tagged as murine	5	256

Test dataset (668 samples)

On the validation dataset:

$$F1_H = 0.9951^{[15]} \quad F1_M = 0.9924^{[16]} \quad Acc = 0.9940, \quad Q = 0.9927$$

And on the test dataset:

$$F1_H = 0.9890^{[17]} \quad F1_M = 0.9827^{[18]} \quad Acc = 0.9865, \quad Q = 0.9850$$

¹⁵Precisely: $F1_H = 0.995061728395062$.

¹⁶Precisely: $F1_M = 0.992366412213740$.

¹⁷Precisely: $F1_H = 0.988957055214724$.

¹⁸Precisely: $F1_M = 0.982725527831094$.

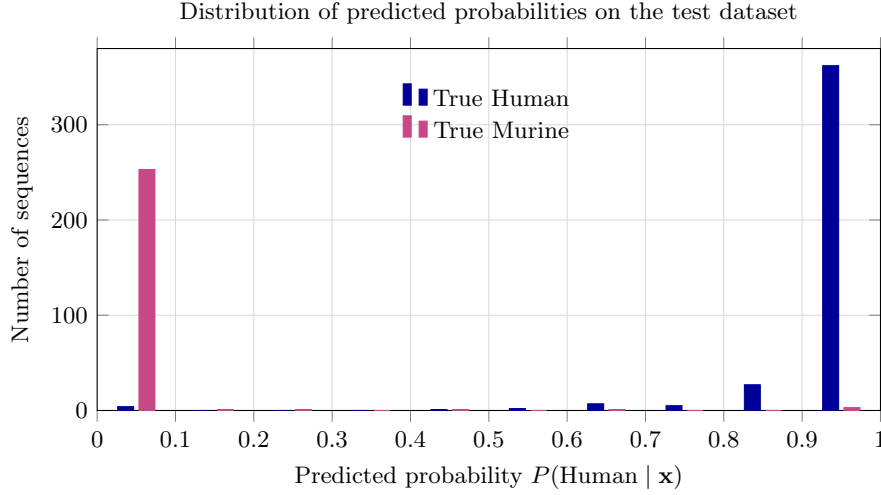


Figure 10: Distribution of predicted probabilities $P(\text{Human} | \mathbf{x})$ on the test dataset (408 human and 260 murine sequences). Most sequences have probabilities close to 0 or 1, with few sequences near 0.5.

7.5 Permutation importance

Because the kernels are correlated, the regression weights are shared among them. To see how much each kernel matters on its own, we compute permutation importance on the test dataset: we shuffle the values of each kernel across the 668 test chains 500 times, and measure the drop in accuracy and F1 scores.

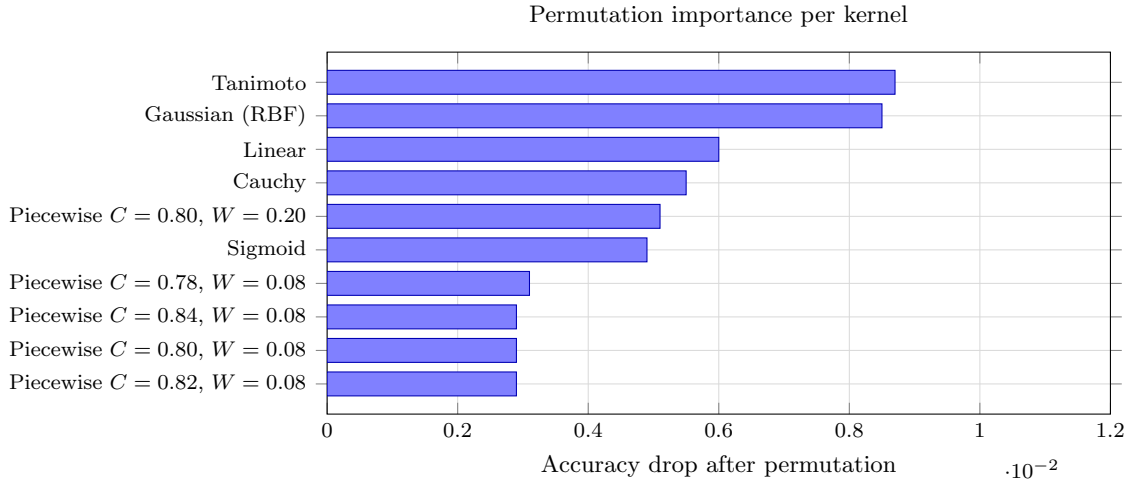


Figure 11: Drop in test accuracy when each kernel is randomly shuffled 500 times. [See appendix for complete table with F1 drops.](#)

Every single kernel causes a drop in accuracy when shuffled, showing that all 10 features help the classifier. Tanimoto and Gaussian have the largest impact.

7.6 Comparison with decision tree (CART)

We also tested a decision tree (CART) trained on the same 10-dimensional vectors. A comparison of the test results shows:

Model	Accuracy	F1 Human	F1 Murine	$\mathcal{Q}$
CART (depth 5)	0.9656	0.9714	0.9568	0.9131
Logistic regression	0.9865	0.9890	0.9827	0.9850

Table 1: Test dataset results comparing the decision tree and logistic regression models.

The logistic regression model reaches an accuracy of 98.65% and an $F1_H$ of 0.9890. Combining continuous outputs linearly gives a smooth separation without the sharp axis-aligned splits of the decision tree.

7.7 Accuracy and certainty frontier

We compare all single kernels and the logistic regression model on the test dataset in Figure 12:^[19]

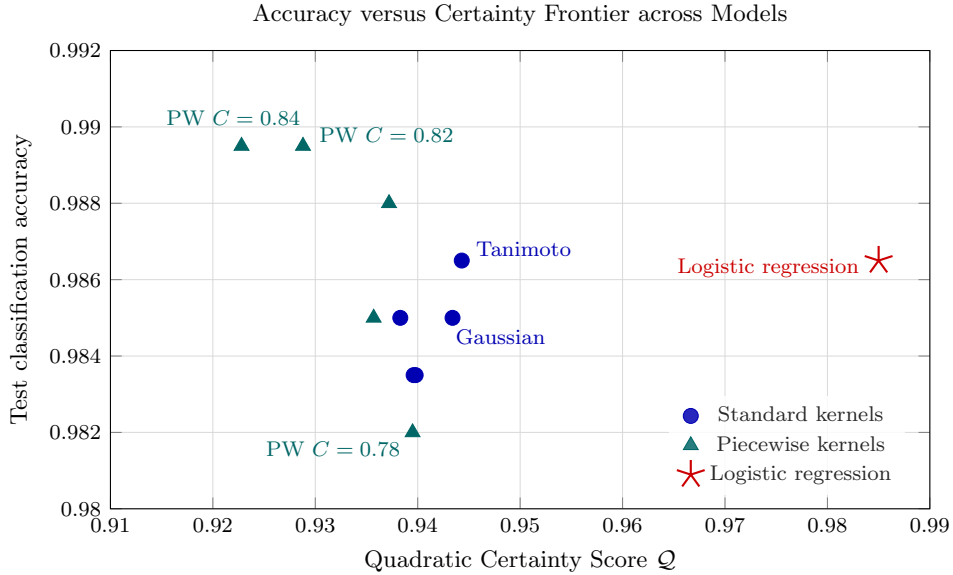


Figure 12: Test classification accuracy versus Quadratic Certainty Score $\mathcal{Q}$ across standard kernels (blue circles), piecewise kernels (teal triangles), and the logistic regression model (red star).

While piecewise kernels reach 98.95% accuracy and standard kernels reach $\mathcal{Q} \approx 0.944$, the logistic regression model reaches $\mathcal{Q} = 0.9850$ and 98.65% accuracy, improving certainty over every individual kernel.

¹⁹CART is not included here because of its lower accuracy and certainty score.

Appendix

SVM formula deduction

We have a cloud of points in a higher-dimensional space and we want a plane that separates the ones marked human and the one marked murine^[20]. But we do not only want *one* that does that, we need the plane that does that the best: that is the one that maximizes the distance to the closest points.

We *must* assume that the two groups of points are separable by a plane in our space and let us say the distance^[21] between the two clouds is $2d$; we will need this later.

Our solution will have the form:

$$\mathbf{w}^T \mathbf{X} + b = 0,$$

where $\mathbf{w}$ is the vector perpendicular to the solution plane, $\mathbf{X}$ is a variable-coordinate vector and b is a real number we will call bias.

Since the norm of $\mathbf{w}$ is arbitrary, we can set $\|\mathbf{w}\| = 1/d$. Thus, any of our data points inputted in the function $\mathbf{x} \mapsto |\mathbf{w}^T \mathbf{x} + b|$ will give an output greater or equal to 1 (the nearest ones will give 1, otherwise strictly greater than 1).^[22]

Furthermore, we know whether the result will be negative and then inverted by the absolute value. The ones such that $\mathbf{w}^T \mathbf{x}_i + b \leq -1$ will have also the negative tag: $s_i = -1$. Thus,

$$(\mathbf{w}^T \mathbf{x}_i + b) s_i \geq 1 \quad \forall i \in \{1, \dots, n\}.$$

Maximizing d is equivalent to minimizing $\|\mathbf{w}\|$, which is again equivalent to minimizing $\frac{1}{2} \|\mathbf{w}\|^2$, this will come handy after some computations.

We have a optimization problem subject to restrictions:

$$\underset{\mathbf{w}, b}{\text{minimize}} \quad \frac{1}{2} \|\mathbf{w}\|^2, \quad \text{subject to } (\mathbf{w}^T \mathbf{x}_i + b) s_i \geq 1 \quad \forall i \in \{1, \dots, n\}$$

We define the Lagrangian with the classification margin:

$$\mathcal{L}(\mathbf{w}, b, \lambda) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \lambda_i (s_i (\mathbf{w}^T \mathbf{x}_i + b) - 1)$$

We set the partial derivatives to 0:

$$\nabla_{\mathbf{w}} \mathcal{L} = 0 \implies \mathbf{w} = \sum_{i=1}^n \lambda_i s_i \mathbf{x}_i \quad \frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{i=1}^n \lambda_i s_i = 0$$

We substitute $\mathbf{w}$ back into the Lagrangian and apply the restriction for b :

$$\mathcal{L}(\lambda_1, \dots, \lambda_n) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j s_i s_j \mathbf{x}_i^T \mathbf{x}_j - \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j s_i s_j \mathbf{x}_i^T \mathbf{x}_j - b \sum_{i=1}^n \lambda_i s_i + \sum_{i=1}^n \lambda_i$$

We simplify to obtain the dual formulation:

$$\mathcal{L}(\lambda_1, \dots, \lambda_n) = \sum_{i=1}^n \lambda_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j s_i s_j \mathbf{x}_i^T \mathbf{x}_j,$$

²⁰Those are what we call tags: +1 for human and -1 for murine chains.

²¹Distance from one set to another is defined as the minimum distance from one point of one set to a point of the other one. Those two (or more) points will be called support vectors.

²²Notice that $\mathbf{x}$ and $\mathbf{x}$ are different, in fact, $\mathbf{x}_i$ is the data point $\mathbf{x}_i$ in the transformed space ($\varphi(\mathbf{x}_i) = \mathbf{x}_i$, being φ the transformation of our space such that $\varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j = k(\mathbf{x}_i, \mathbf{x}_j)$).

We use our kernel notation for the inner products:

$$\mathcal{L}(\lambda_1, \dots, \lambda_n) = \sum_{i=1}^n \lambda_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j s_i s_j k(\mathbf{x}_i, \mathbf{x}_j),$$

$$\text{with } \sum_{i=1}^n \lambda_i s_i = 0 \text{ and } \lambda_i \geq 0 \ \forall i \in \{1, \dots, n\}$$

This way we find the lagrange multipliers, maximizing the Lagrangian subject to the restrictions given.

For our decision function we just want to know which side of the plane is on. We achieve that plugging the new data point into our plane equation:

$$\mathbf{w}^T \mathbf{x} + b = \left(\sum_{i=1}^n \lambda_i s_i \mathbf{x}_i^T \right) \mathbf{x} + b = \sum_{i=1}^n \lambda_i s_i \mathbf{x}_i^T \mathbf{x} + b = \sum_{i=1}^n \lambda_i s_i k(\mathbf{x}_i, \mathbf{x}) + b$$

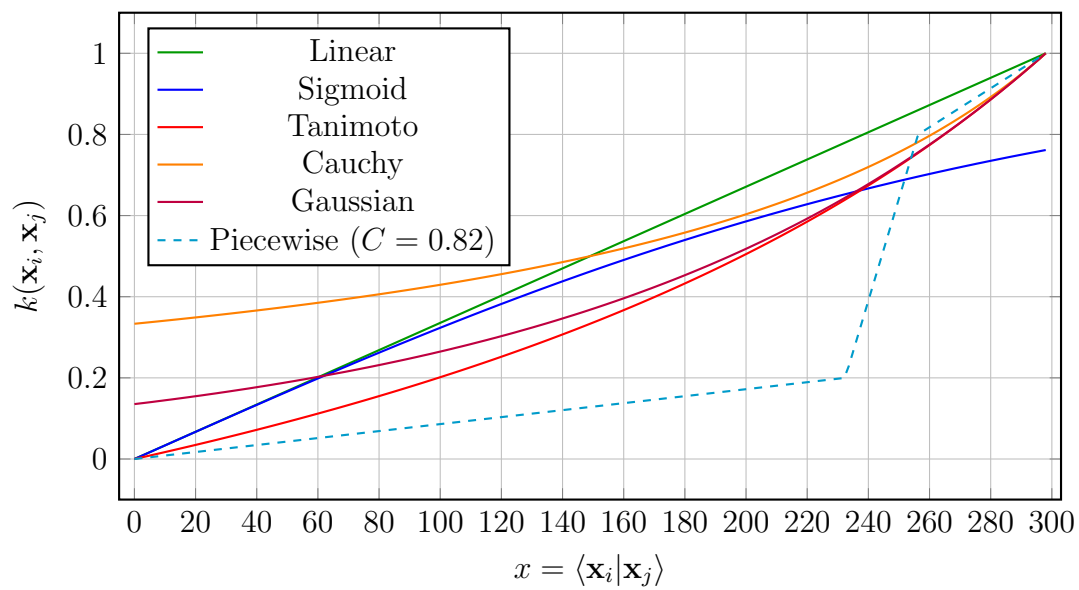
$$f : \mathbf{x} \mapsto b + \sum_{i=1}^n \lambda_i s_i k(\mathbf{x}_i, \mathbf{x})$$

Also, since $(\mathbf{w}^T \mathbf{x}_p + b)s_p = 1$ for some support vector $\mathbf{x}_p$, expanding $\mathbf{w}$:

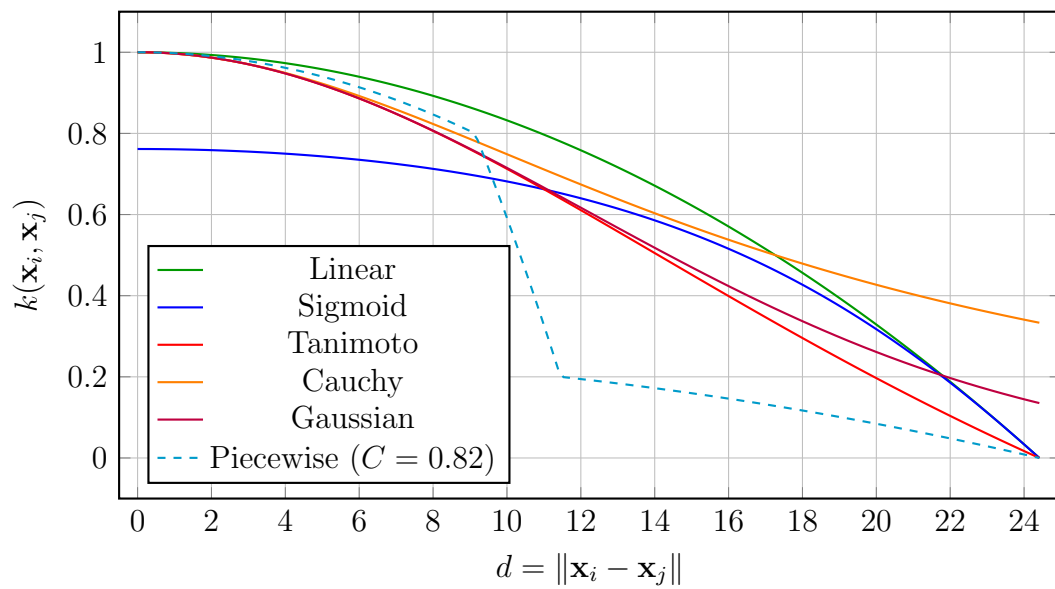
$$b = s_p - \sum_{i=1}^n \lambda_i s_i k(\mathbf{x}_i, \mathbf{x}_p)$$

SVM kernel function comparison

All Kernels vs. Inner Product



All Kernels vs. Distance



Piecewise kernels performance

Kernel parameters	Accuracy	F1 _H	F1 _M	$\mathcal{Q}$
Piecewise ($C = 0.80$, $W = 0.20$)	0.9880	0.9903	0.9844	0.9372
Piecewise ($C = 0.80$, $W = 0.08$)	0.9850	0.9878	0.9806	0.9357
Piecewise ($C = 0.82$, $W = 0.08$)	0.9895	0.9915	0.9865	0.9288
Piecewise ($C = 0.84$, $W = 0.08$)	0.9895	0.9915	0.9865	0.9228
Piecewise ($C = 0.78$, $W = 0.08$)	0.9820	0.9853	0.9769	0.9395

Table 2: Performance of the custom piecewise kernels on the test dataset (668 samples), including the Quadratic Certainty Score $\mathcal{Q}$.

Permutation importance

Kernel	Accuracy drop	F1 _H drop	F1 _M drop
Tanimoto	+0.0087	+0.0072	+0.0109
Gaussian (RBF)	+0.0085	+0.0070	+0.0106
Linear	+0.0060	+0.0050	+0.0075
Cauchy	+0.0055	+0.0046	+0.0069
Piecewise ($C = 0.80$, $W = 0.20$)	+0.0051	+0.0042	+0.0064
Sigmoid	+0.0049	+0.0041	+0.0061
Piecewise ($C = 0.78$, $W = 0.08$)	+0.0031	+0.0026	+0.0039
Piecewise ($C = 0.84$, $W = 0.08$)	+0.0029	+0.0024	+0.0037
Piecewise ($C = 0.80$, $W = 0.08$)	+0.0029	+0.0024	+0.0036
Piecewise ($C = 0.82$, $W = 0.08$)	+0.0029	+0.0024	+0.0036

Table 3: Permutation importance measured on the test dataset across 500 random shuffles per feature.

Complete ten-kernel correlation heatmap

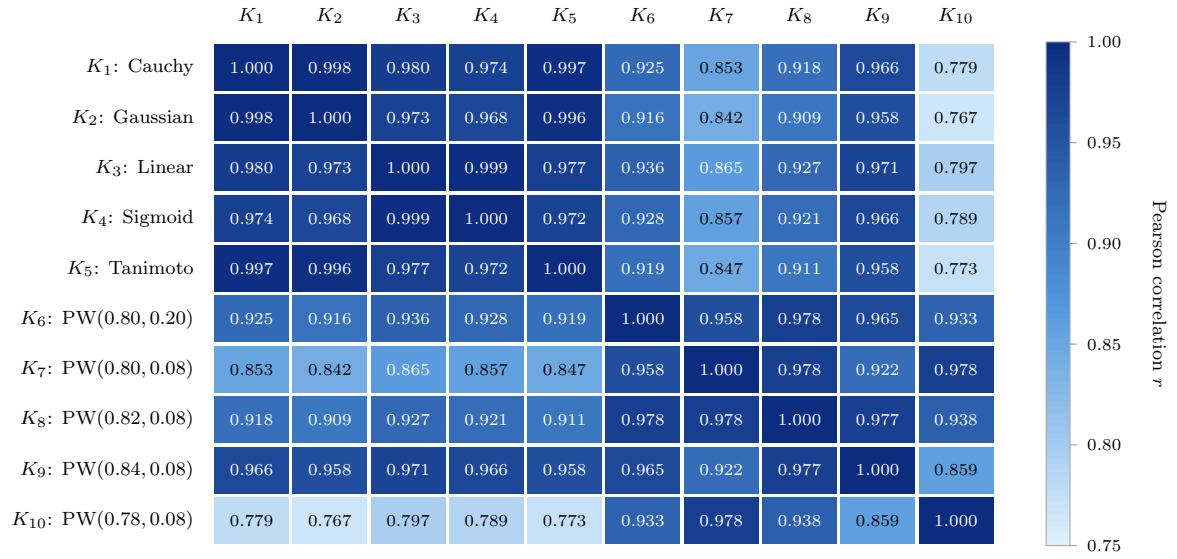


Figure 13: Correlation heatmap across the decision function outputs of all ten evaluated standard and custom piecewise kernels on the test dataset (668 sequences).